Homotopy

$f, g: X \to Y$ continuous

if $\exists$ $H: X \times I \to Y$ cont. (I = time)

such that
$H(x,0) = f(x), \ \forall x \in X$
$H(x,1) = g(x), \ \forall x \in X$

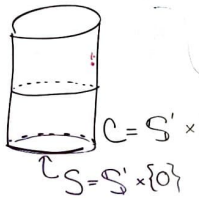
then, $f \simeq g$.
"f is homotopic to g"

X, Y topo spaces

$X \simeq Y$ means $\exists$ $X \underset{g}{\overset{f}{\rightleftarrows}} Y$

Such that $f \circ g \simeq id_Y$
$g \circ f \simeq id_X$

ex:



retracts onto S

Why?

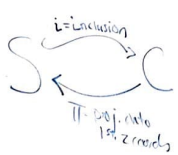
$H(c_1, c_2, c_3, t) = (c_1, c_2, t c_3)$

or

$H(c, t) = t \cdot \pi(c) + (1-t)\, i(c)$

Claim: $C \simeq S$

Proof: (in groups)



Can you:
1) show i is continuous?
2) show π is continuous?
3) show $f \simeq f$?
4) show H is continuous?

$i \circ \pi = \pi \simeq id_C$

$\pi \circ i = id_S$

homotopy defines an equivalence relation on topological spaces

... but ...

not very computable.

Group

A group $(G, +)$ is a set G with an binary operation $+$ such that

1) closure: $\forall g_1, g_2 \in G,\ g_1 + g_2 \in G$
2) identity elt. $\exists e \in G$ such that $\forall g \in G,\ g + e = e + g = g$
3) inverse: $\forall g \in G,\ \exists h \in G$ such that $g + h = e = h + g$
4) associativity: $\forall g_1, g_2, g_3 \in G,\ (g_1 + g_2) + g_3 = g_1 + (g_2 + g_3)$

$(G, +)$ is abelian if also:

5) commutative: $\forall g_1, g_2 \in G,\ g_1 + g_2 = g_2 + g_1$

examples:

- $(\mathbb{R}, +)$, $e = 0$
- $(\mathbb{R}, \cdot)$, $e = 1$

Conventions:

- $a \cdot b = ab$
- e in $(*, +)$ written as 0
- e in $(*, \cdot)$ written as 1

Field

$(G, +, \cdot)$ is a field if

(1) $(G, +)$ is an abelian group (id is called 0)
(2) $(G \setminus \{0\}, \cdot)$ is an abelian group (id is called 1)
(3) mult. distributes over addition
$\forall g_1, g_2, g_3 \in G$

$$g_1 \cdot (g_2 + g_3) = g_1 \cdot g_2 + g_1 \cdot g_3$$

Questions:

Which of these are fields/groups?

1) $(\mathbb{R}, +, \cdot)$
2) $(\mathbb{Z}, +, \cdot)$
3) p = prime number (e.g., $p = 2$)

$\mathbb{Z}_p := \mathbb{Z} \bmod p$

$= \{z \in \mathbb{Z} \mid z \sim z' \text{ iff } z \equiv z' \bmod p\}$

$\left(\cong \mathbb{Z}/p\mathbb{Z} \text{ (quotient group)}\right)$

4) $\mathbb{Z}_4$

...e fields/groups?

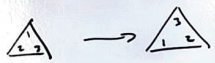
...r (e.g., p=2)

...z' iff
...z≡z' mod p}
...tient group))

5) Dihedral Group

Given: △ equilateral,

G = operations we can do to map △→△

rotate: 

flip: 

(generalizes to a regular polygon)

6) Permutation group